

ChE319 Transport Phenomena

Spring 2026

“Transport phenomena” is an introduction to the physics of fluid, heat and mass transfer in chemical engineering processes.

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TA:

Office Hours:

Lectures:

MWF 1:00 – 2:00 PM CPE 2.216

Please be punctual. Attendance is not compulsory, but is strongly recommended.

Recitation:

W 4:00PM- 6:00PM CPE 2.216.

Please be punctual. Homework is due at the beginning of the class. Most of the session will be used to discuss problems similar to the homework. In the last 40 minutes, weekly quiz will be conducted.

Textbook: “Transport Phenomena”, 2nd Edition, by Bird, Stewart, and Lightfoot.

Final Grade

Pre-Lecture Problems	10% (Every Lecture)
Homework	5%
Quizzes (Both recitation and CANVAS)	20% (will drop lowest quiz grade)
Midterm Exams	50%
Final Exam	15%

Final letter grade will be given based on Class average. The class average for this course is B-. There are two requirements for a passing grade C-: You should have an exam grade of 60 points and have a course average that will be determined from your class average and standard deviation.

Homework: Homework is due at the beginning of class. Late homework will not be accepted. Homework must be neat and presented in the standard ChE format.

Quizzes: There will be a quiz during all recitation sessions. The quiz will be based on material covered during class, homework and recitation. All quizzes count and the weighing of a quiz depends on the difficulty of the problem. All quizzes are closed book and notes.

Exams:

There will be three exams during the semester. Exams are closed book and closed notes.

Exam – I	Feb 6 th	(Evening 6 to 9 pm CPE 2.216, 2.220)
Exam – II	Mar 6 th	(Evening 6 to 9 pm CPE 2.216, 2.220)
Exam – III	April 10 th	(Evening 6 to 9 pm CPE 2.216, 2.220)
Final Exam	May 1 st	(1:00 pm – 3:00 pm CPE2.220)

(You have one week after the exam is returned to you to request for re-grading. Re-grading requests should be submitted in writing at the end of that lecture period or during office hour)

Absence: No excuse for missed tests, exams, or quizzes will be accepted other than written certified medical excuses or written letters on university letterhead for UT-related school activities.

Note: Laptop computers, mobile phones and other electronic communications devices are NOT allowed during recitations or lectures.

Course content

- Review of Mathematical tools
- Differential equations
- Coordinate systems, Multivariable integration
- Transport by molecular motion
 - Newton's Law of viscosity
 - Fourier's Law of heat conduction
 - Fick's Law of binary diffusion
 - Temperature and pressure dependence of transport coefficients
- Transport in one dimension (Shell Balance method)
- Couette flow (between parallel plates and annular region)
- Flow of a falling film (gravitational flow)
- Flow through conduits (Circular, rectangular, annular, etc., Hagen-Poiseuille equation, Laminar flow, significant of Reynolds number, Newtonian fluids, average velocity,)
- Heat conduction through composite walls (convection boundary conditions, series and parallel resistance, Log mean temperature difference,
- Heat conduction with a heat source (Critical insulation thickness)
- Heat conduction through composite walls, cooling fin
- Diffusion through stagnant gas film
- Diffusion with chemical reaction
- General transport equations
- Equation of continuity
- Equation of motion
- Equation of mechanical energy
- Energy equation
- Application of equations of change to various physical problems in different coordinates
- Transport with two independent variables (unsteady state problems)
- Flow near wall suddenly set in motion
- Unsteady heat conduction in solids
- Lumped models (Reduction in dimensionality)
- Transport in turbulent flow
- Transport across phase boundaries

- Friction factor, heat and mass transfer coefficient
- Creeping flow around immersed objects
- Flow through packed columns
- Heat transfer coefficient for Forced convection
- Transport in large systems
- Application of Bernoulli Equations to simple flow problems.